

NATIONAL INCOME

INTRODUCTION- This is a term used interchangeably with National dividend, national output and national expenditure.

Thus NI means the **total value of goods and services produced annually in a country**. It includes payments made to all resources in the form of wages, interest rent and profits.

NI measures the money value of the flow of output of goods and services produced within an economy over a period of time.

The information which it gives is important to measure the level and rate of economic growth of NI(y)

It helps to consider;

1. The **rate of economic growth**
2. **Changes over time to the average living standard of the production.**
3. Changes over time to the distribution of income between different groups within the population(**scale of income and wealth inequalities within a society**)

Real Concepts of national income

1. G.D.P – **money value of all goods and services produced annually** in a country using **exclusively the resources of a country.**
2. G.N.P. – The **national total production of goods and services** for one year evaluated in terms of the **market price** of goods and services produced.

It includes all the economics production in the economy during one year and all the factors incomes from abroad.

$G.N.P. = G.D.P + \text{net factor incomes from abroad (total exports- total imports)}$

3. N.N.P—it is the production of goods and services in a country during one year **taking into accounts the value of capital consumed (depreciation)**

$N.N.P = G.N.P - \text{Depreciation}$ or it is the NI at market prices.

Or it is the national income at market prices.

4. N.N.I—it's the NI at factor cost. It is the total of all income payments received by the factor input i.e land, labour, capital and enterprises and organization during one year.

$NNI = NNP - \text{indirect taxes} + \text{subsidies}$

It tells as how the National output is distributed amongst the various factor inputs in line of the services rendered to them.

5. Personal income—Total income that is actually available for distribution amongst the individuals or households in a country during a year.

$PI = NNI - \text{corporation tax} - \text{retained profit} + \text{contribution to social security scheme} + \text{transfer of payments} + \text{benefit of social security scheme}.$

6. D.P.I--this is a part of PI that is left behind after the payment of personal direct taxes i.e the income of the individual available for consumption.

$D.P.I = PI - \text{personal direct taxes}$

7. Per capital income—it's the income of the year's distribution per unit of production

$P.C.I = \frac{NNI \text{ for a given year}}{\text{Population during the year}}$

1. Real net National income of a country for a given year expressed in terms of the constant prices or the prices prevailing in the base year.

Measuring National income

We use national income accounts to measure

1. The rate of economic growth
2. Change over time to the average living standards of the production
3. Changes over time to the distribution of income between different groups within the production

GDP—measures the value of the output produced within the domestic boundaries of the economy over a given period – though GDP includes the output of the foreign owned business that are located in Kenya following foreign direct investment in an economy.

Methods of measuring national income

There are three ways of measuring GDP and all of which should sum to the same amount.

1. Output— Total value of the output of goods and services produced in an economy.
2. Spending— Total amount of expenditure taking place in the economy.
3. Incomes— Total income generated through production of goods and services.

1. Expenditure Method

This approach measures total spending on goods and services in the economy. It includes:

- **Consumption (C)** – Spending by households
- **Investment (I)** – Spending by businesses on capital goods
- **Government expenditure (G)** – Spending by the government on public services
- **Net exports (X - M)** – The difference between exports and imports

Formula:

$$\text{GDP} = C + I + G + (X - M)$$

National output = National expenditure = N.I (aggregate demand)

- 1) Expenditure method—the sum of spending on an economy produced goods and services measured at current market prices i.e $= C + I + G + (X - M)$

Where; C-household spending

I—capital investment spending

G—government spending

X—exports of goods and services

M—import of goods and service

Numerical example

Given these figures calculate the National income using expenditure method.

Component	Expenditure (KES Million)
Household Consumption (C)	700
Business Investments (I)	300
Government Spending (G)	400
Exports (X)	200
Imports (M)	100

Solutions

$$\text{GDP} = C + I + G + (X - M)$$

$$\text{GDP} = 700 + 300 + 400 + (200 - 100)$$

$$\text{GDP} = 700 + 300 + 400 + 100 = 1,500 \text{ million KES}$$

2. Income Method

This approach **sums all incomes earned by individuals and firms in the economy**. It includes:

- **Wages and salaries** (earned by employees)
- **Profits of businesses** (after taxes)
- **Rent** (from land and property)
- **Interest** (earned on capital)

Formula:

$$\text{National Income} = \text{Wages} + \text{Profits} + \text{Rent} + \text{Interest}$$

Income method (sum of factor income)—this is the sum of the incomes earned through the production of goods and services i.e

- A. Income from employed self-people
- B. Profits of private sector companies
- C. Rent income from land

We also include; transfer of payment—state pension paid to retired people or other welfare assistance to child/household benefits, private transfer of money from one individual to another. Income that is not registered with custom excise collection body i.e shadow economy.

Numerical example

Support an economy is characterized by the following information.

Component	Income (KES Million)
Wages and Salaries	600
Profits of Businesses	400
Rent from Property	200
Interest on Capital	150

Calculate National income using income method

Solutions

$GDP = \text{Wages} + \text{Profits} + \text{Rent} + \text{Interest}$

$GDP = 600 + 400 + 200 + 150$

$GDP = 1,350$ million KES

The answer will be 1350 million Kenya shillings

3. Output method

Under this we use the concept of value added.

This approach measures the total value of goods and services produced within a country in a given period (usually a year). It includes:

- Agriculture and primary sector output (e.g., farming, fishing, mining)
- Manufacturing and industrial production
- Service sector output (e.g., banking, education, healthcare)

Formula:

$$GDP = \sum (\text{Value of Output} - \text{Intermediate Goods})$$

This method avoids double counting by only considering the final value of goods and services.

This is a measure of GDP by adding together the value of output produced by each of the production sector in the economy using the concept of value added i.e the increase in the value of the product at each successful stage of the production process.

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- **Agriculture and primary sector output** (e.g., farming, fishing, mining)
- **Manufacturing and industrial production**
- **Service sector output** (banking, education, healthcare)

Formula:

$$\text{GDP} = \sum (\text{Value of Output} - \text{Intermediate Goods})$$

It is used to avoid the problem of double counting problem of intermediate inputs

Numerical example

Sector	Total Output (Ksh Million)	Intermediate Goods (Ksh Million)	Final Output in Ksh Million) (total output – intermediate goods)
Agriculture	500	100	400
Manufacturing	800	300	500
Services	600	100	500

Total GDP using Output Method:

$$\text{GDP} = \sum (\text{Value of Output} - \text{Intermediate Goods})$$

$$\text{GDP} = (500 - 100) + (800 - 300) + (600 - 100)$$

$$\text{GDP} = 400 + 500 + 500 = 1,400 \text{ million}$$

Factors Determining the Size of an Economy

1. **Natural Resources** – Availability of minerals, fertile land, water, and energy sources boosts production.
2. **Human Capital** – Skilled, educated, and healthy workers enhance productivity and economic growth.
3. **Capital Formation** – Investment in infrastructure, machinery, and technology increases output.

4. **Technology and Innovation** – Advanced technology improves efficiency and competitiveness.
5. **Government Policies** – Stable policies on taxation, trade, and investment encourage growth.
6. **Political Stability and Governance** – A stable government attracts investment and fosters economic progress.
7. **Trade and Market Size** – Access to large markets and international trade expands economic opportunities.
8. **Industrialization and Manufacturing** – Strong industries create jobs and boost national income.
9. **Monetary and Fiscal Policies** – Effective management of inflation, interest rates, and public spending supports stability.
10. **Entrepreneurship and Business Environment** – A favorable environment for businesses stimulates economic activity.

Difficulties in Measuring National Income

1. **Non-Market Activities** – Household work, volunteer services, and subsistence farming are not recorded.
2. **Informal and Underground Economy** – Black market activities, tax evasion, and informal businesses go unreported.
3. **Double Counting** – Including intermediate goods along with final goods leads to inflated GDP figures.
4. **Difficulty in Valuing Services** – Public services like education and healthcare lack clear market prices.
5. **Price Fluctuations and Inflation** – Changes in price levels affect real GDP comparisons over time.
6. **Data Collection and Accuracy Issues** – Poor record-keeping, lack of proper record keeping

Uses of Nation income accounting

1. **Economic Planning and Policy Formulation** – Helps governments design fiscal and monetary policies.
2. **Measuring Economic Growth and Development** – Tracks GDP trends to assess economic progress.
3. **International Comparisons** – Compares economic performance between countries.
4. **Standard of Living Assessment** – Evaluates income levels and quality of life using GDP per capita.
5. **Business and Investment Decisions** – Helps businesses and investors analyze economic trends.
6. **Employment and Wage Analysis** – Assists in tracking job creation, unemployment, and wage policies.
7. **Taxation and Government Revenue Planning** – Aids in estimating tax revenues and budget allocations.

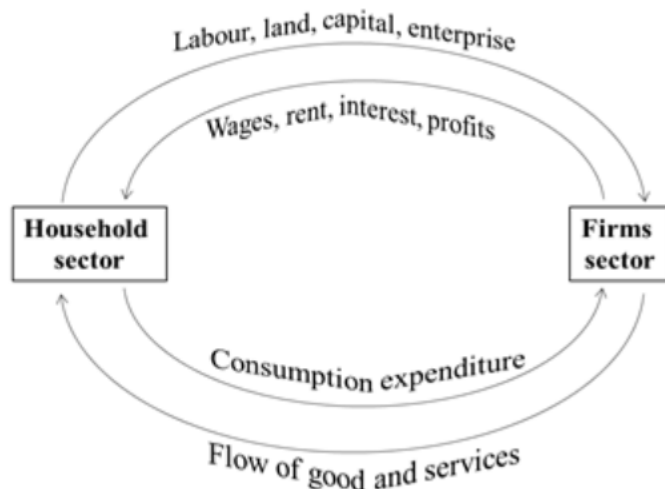
THE CIRCULAR FLOW OF INCOME AND EXPENDITURE

This refers to the process whereby the N.I and expenditure of an economy flow in a circular manner continuously through time.

Incomes from savings, investments, taxation (which is taken as leakage from the circular flow), government expenditure (an injection into the circular flow) and exports and imports are what show is in a simple closed economy.

Simple to Sector Economy

Lets us take a case of a simple economy with two sector that's household sector and business



sector

Figure 1: Circular flow of income and expenditure in a two sector economy

The two sector economy household owns all factors of production, is land, labour and capital. This sector receives income by selling the services of these factors to the business sector

The household sector buys the output of product of the business sector as shown in the figure above.

The business sector, in turn, makes payments to the households for capital supplied e.t.c. thus payments go round in a circular manner from the business sector to the household sector, and from household sector to business sector.

Circular Flow of Income in a Three Sector Economy

To this we add the government sector so as to make it three closed sector economy model circular flow of income and expenditure. For this we add taxation and government purchase (or expenditure) in our presentation

Taxation is a leakage and purchases are injections in the circular flow.

Taxes in the form of personal income tax and commodity taxes paid by the household sectors are outflows or leakage to the circular flow

In return the government makes purchases of the household services, makes transfer of payments in the form of old age pensions, unemployment relief, sickness benefits etc. and also spends on them to provide certain social services like education, health, housing, water, parks and other facilities. All such expenditure by the government are injections into the circular flow

To business sector and the government sector. All taxes paid by the business sector to the government and leakages and government purchases to the business are injections to the circular flow.

Taking all the three taxes are leakages to the circular flow of income. It tends to reduce consumption and savings of the household sector. Reduced consumption and savings of household in turn, reduces the sales income of firms. On the other hand taxes on business sector tends to reduce their investment and production.

The government offsets these leakages by making purchases from the business sector and buying services from the household sector to the amount of taxes. Thus total sales again equals production of firms in this way, the circular flows of income and expenditure remain in equilibrium.

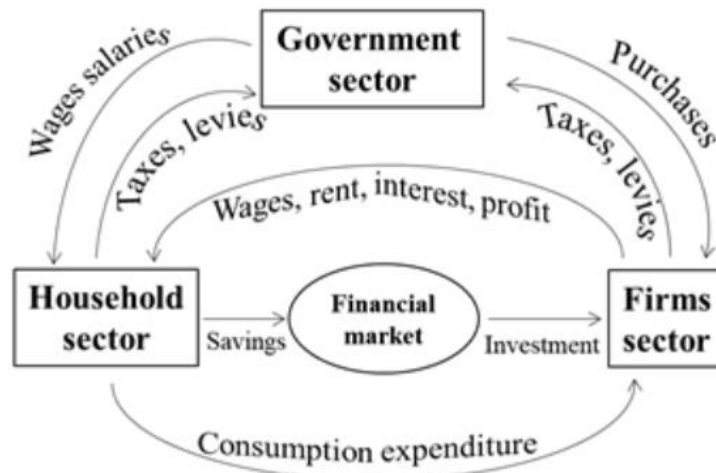


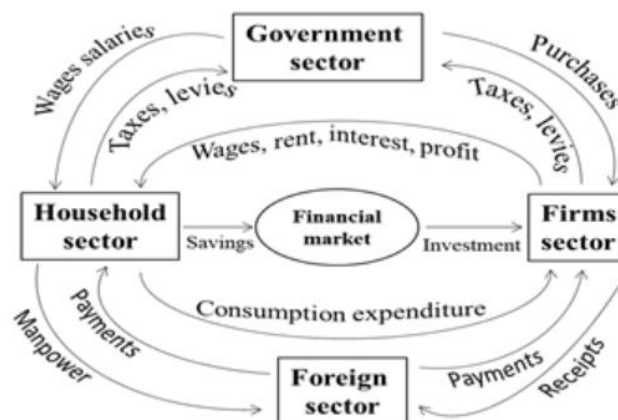
Figure 2: Circular flow of income and expenditure in a three sector economy

The government makes investments and for this purchase of goods from firms and also factors of production from household. Thus these are injections to the circular flow of income and taxes are leakages.

The Circular Flow in a Four Sector Economy

So far we have had a close economy but real sense it is an open economy where foreign trade plays an important role. Exports are injections to circular flow of income for they buy goods and pay for the goods and pay for the goods and services.

On the other hand imports are leakages from the circular flow of income. They are expenditures incurred by the household sector to purchase goods from foreign countries.



Take the inflows and outflows of the household business and government sector in relation to the foreign sector. The household sector buys goods important from abroad and makes payments for them which is a leakage from the circular flow. The household may receive transfer of payments from foreign sector.

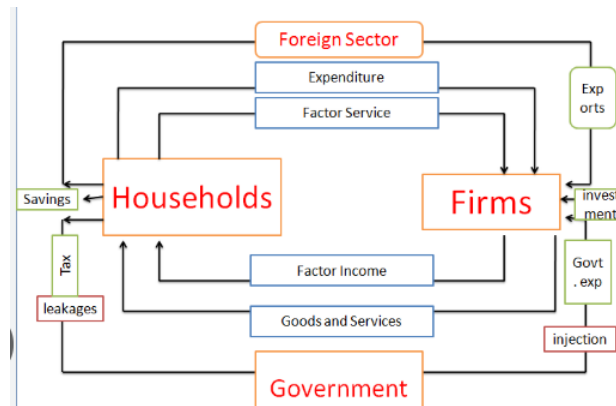


Figure 3: Circular flow of income and expenditure in a four sector economy

Injections (Inflows) and

In the real economy inflows and leakages occur in the expenditure and income flows such leakages are saving and inflows or injections are investments which equal to each other.

Under this, expenditure has now two alternative paths from household and business sector

1. Directly via consumption expenditure
2. Indirectly via investment expenditure.

Injections are the additional sources of money entering the economy, increasing the flow of income. The three main injections are:

a) Investment (I)

- When businesses **invest in capital goods**, they increase production and employment.
- Example: A company invests KES 200,000 in new machinery, increasing production capacity.

b) Government Spending (G)

- When the **government spends on infrastructure, wages, and services**, it boosts economic activity.

- Example: If the government spends KES 300,000 on building roads, workers get paid and spend their income.

c) Exports (X)

- When a country **sells goods and services to foreign markets**, money enters the economy.
- Example: If a business exports goods worth KES 100,000, foreign buyers bring money into the domestic economy.

Total Injections Formula:

$$\text{Injections} = I + G + X$$

Withdrawals (Leakages)

Leakages are the portions of income that are not spent on domestic goods and services, reducing the flow of income. The three main types of leakages are:

a) Savings (S)

- When households or businesses **save money instead of spending it**, this reduces demand for goods and services.
- Example: A household earns KES 100,000 but saves KES 20,000, reducing spending in the economy.

b) Taxes (T)

- When the government **collects taxes**, it withdraws money from households and businesses.
- Example: If the government collects KES 30,000 in taxes from an individual's income, that money is not directly used for consumption or investment.

c) Imports (M)

- When a country **spends on foreign goods and services**, money leaves the domestic economy.
- Example: A business imports machines worth KES 50,000, which sends money abroad rather than circulating it locally.

Total Leakages Formula:

$$\text{Leakages} = S + T + M$$